

SOLVENCYINSIGHT

AI-Powered Insolvency Prevention Documentation

Member	Role	Work
Rohan	ML/Backend	XGBoost,SHAP,FastAPI
Jonathan	ML/Backend	Training,APIs
Neha	Data/Frontend	DataPrep,React
Mariya	Frontend	UI/UX

1. OVERVIEW

Problem: Insolvency devastates stakeholders. Solution: XGBoost ML predicts bankruptcy 0-100% with SHAP explanations.

2. DATA (NEHA)

100 companies (80 healthy, 20 bankrupt), 500 employees, 20 test companies, 5780 test employees. 12 financial ratios.

3. ML (ROHAN+JONATHAN)

XGBoost: $n_estimators=50$, $max_depth=3$. Altman $Z=1.2X_1+1.4X_2+3.3X_3+0.6X_4+1.0X_5$. SHAP shows top 10 factors.

4. BACKEND (ROHAN+JONATHAN)

FastAPI endpoints: /api/health, /api/financial/upload-single, /api/employee/simulate-layoff

5. FRONTEND (NEHA+MARIYA)

React+TypeScript+Vite+Tailwind. Pages: Dashboard, InsolvencyAnalysis, EmployeeScoring, LayoffSimulation

6. METRICS

Model	Accuracy	AUC
Insolvency	100%	100%
Employee	89.4%	92.7%

SAFE=8% risk, AT_RISK=92% risk

7. RUN

```
pip install -r requirements.txt; python scripts/train_models.py; uvicorn app.main:app --port 8001; npm run dev --port 5180
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